

Technical Document #266: Software Engineering - Comprehensive Overview

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Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Key Takeaways:

- Continuous deployment (CD) automatically deploys code changes to production.
- Code review examines code changes before merging.
- Design patterns provide reusable solutions to common software design problems.